

EGDF21 - Diploma in Infocomm & Media Engineering

EGL218 Data Analysis and Visualisation

Topic 7 : Linear Regression

Objectives:

At the end of the lab session you should know how to:

- a) Explain the line of best fit and linear correlation between two variables
- b) Find the correlation coefficient and the equation of regression line
- c) Interpret the output from simple linear regression analysis and find the predicted values

1 Introduction to Regression Analysis

Regression analysis is used to:

- Predict the value of a dependent variable based on the value of at least one independent variable
- Explain the impact of changes in an independent variable on the dependent variable

Dependent variable: the variable we wish to predict or explain

Independent variable: the variable used to explain the dependent variable

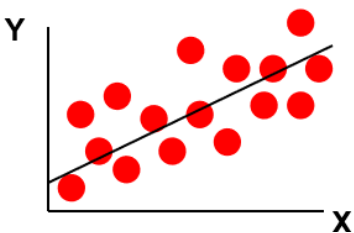
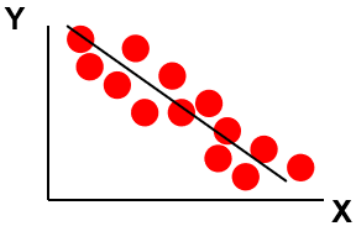
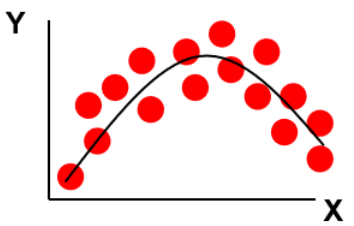
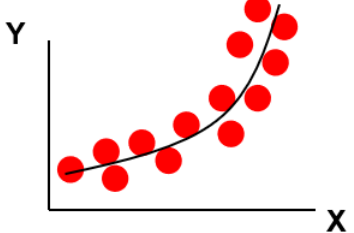
Some examples of dependent and independent variables:

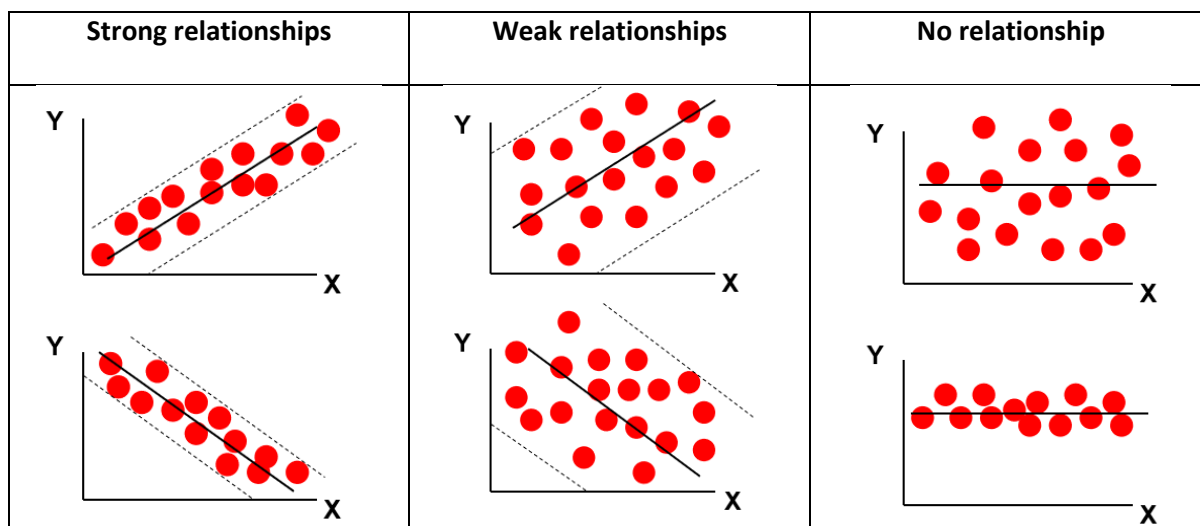
Dependent variables Y	Independent variables X
Height	Weight
Blood pressure	Age
Sales of cold drinks	Climate temperature
Final exam score	Number of lessons missed

2 Simple Linear Regression Model

- Only **one** independent variable, X
- Relationship between X and Y is described by a linear function
- Changes in Y are assumed to be caused by changes in X

2.1 Types of relationships

Linear relationships	Curvilinear relationships
Positive relationship if X increases, Y increases  Negative relationship if X increases, Y decreases 	 



2.2 Correlation Coefficient

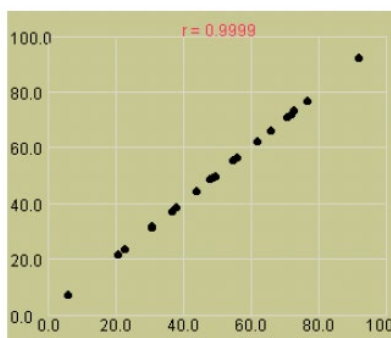
From a scatter diagram, it is often difficult to use our eyes to determine whether there's indeed a linear relationship between the variables.

We can use precise mathematical calculation to help us determine the **degree of linearity** in the relationship between two variables. This is done through the **Pearson product moment correlation coefficient, r** .

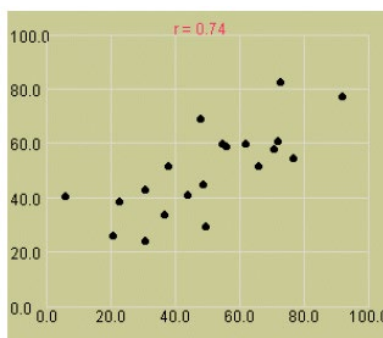
How correlation coefficient, r indicates the linear relationship between two variables?

$-1 \leq r \leq 1$		
Strength of linear relationship	Positive	Negative
Perfect	$r = 1$	$r = -1$
Very strong	$0.8 \leq r < 1$	$-1 < r \leq -0.8$
Strong	$0.4 \leq r < 0.8$	$-0.8 < r \leq -0.4$
Weak	$0.2 \leq r < 0.4$	$-0.4 < r \leq -0.2$
Little / no relationship	$0 \leq r < 0.2$	$-0.2 < r \leq 0$

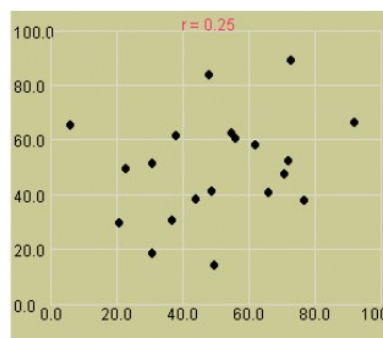
The examples below show the shape of various scatter diagrams and r values.



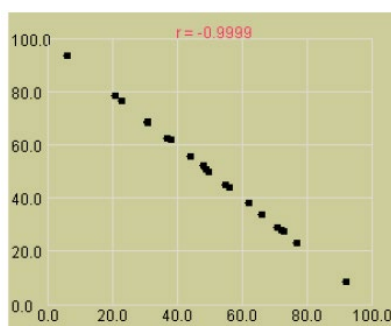
$r = 0.9999$
Very strong,
positive correlation



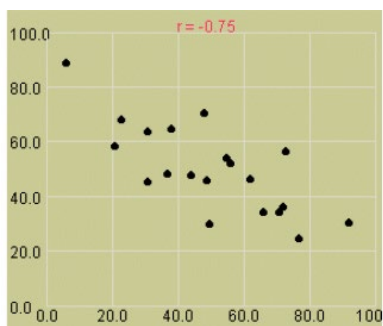
$r = 0.75$
Strong, positive correlation



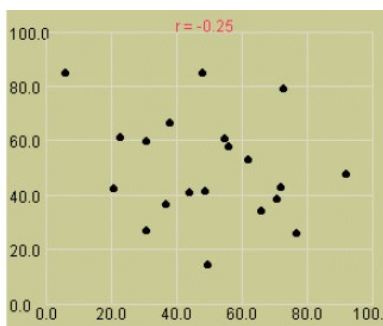
$r = 0.25$
Weak, positive correlation



$r = -0.9999$
Very strong,
negative correlation



$r = -0.75$
Strong, negative correlation



$r = -0.25$
Weak, negative correlation

2.3 Regression Line or Line of Best Fit

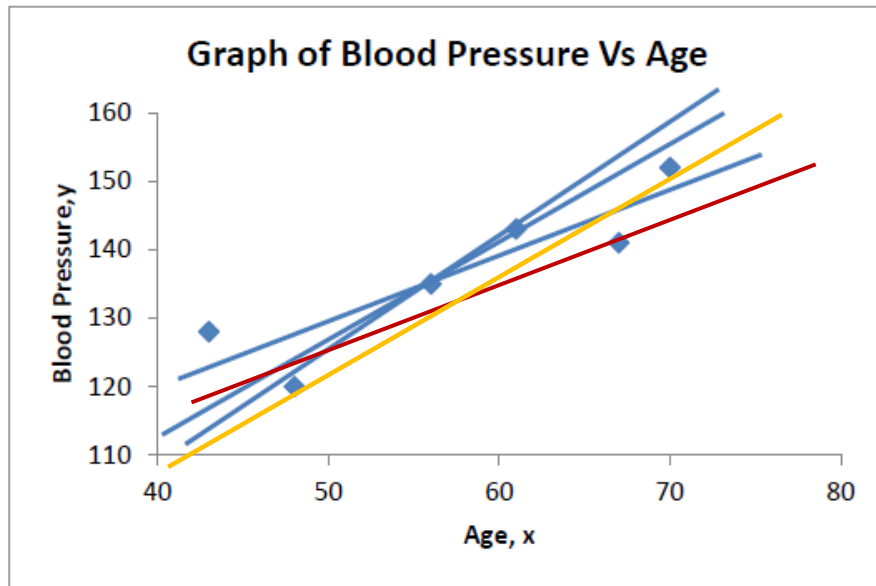
- the line that best fits the given data
- Equation of line : $y = \underline{m}x + \underline{c}$

Slope of line, coefficient of x

y-intercept

Use to answer question such as :

Given any value of x, how can we estimate or predict the value of y?



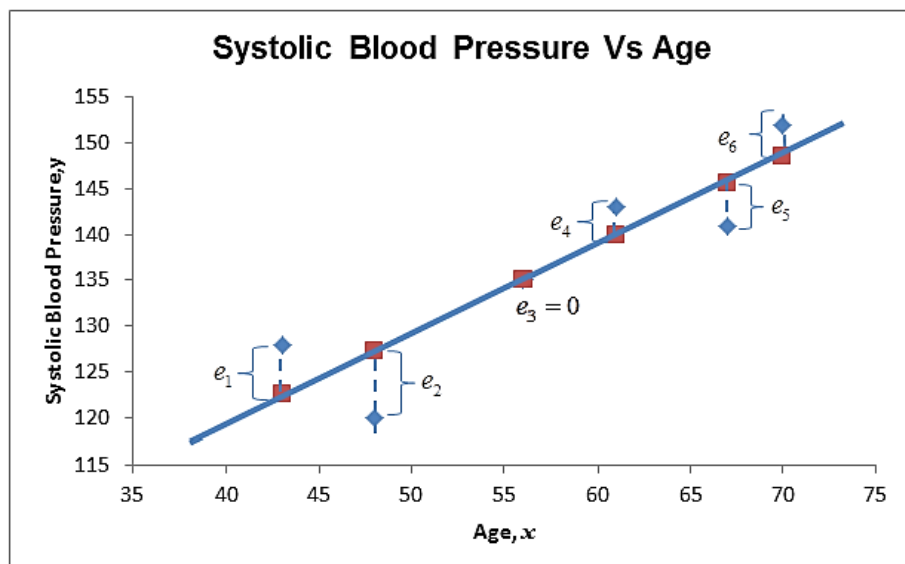
The blue dots on the graph represent the actual observed values of the variables, while the values on the lines are just estimations.

Which is the line of best fit?

2.4 Least square errors

Error or residual of the estimate, $e = y - \hat{y}$

- Difference between the observed y value and the estimate value $\hat{y}$



The line of best fit is the line where the **sum of the square of error** or **SSE**, $\sum_{r=1}^n e_r^2$ is at its **minimum**.

Example 1

A real estate agent wishes to examine the relationship between the selling price of a home and its size (measured in square feet)

A random sample of 10 houses is selected

- Dependent variable (Y) = house price in \$1000s
- Independent variable (X) = square feet

Sample Data for House Price Model

Square Feet (X)	House Price in \$1000s (Y)
1400	245
1600	312
1700	279
1875	308
1100	199
1550	219
2350	405
2450	324
1425	319
1700	255

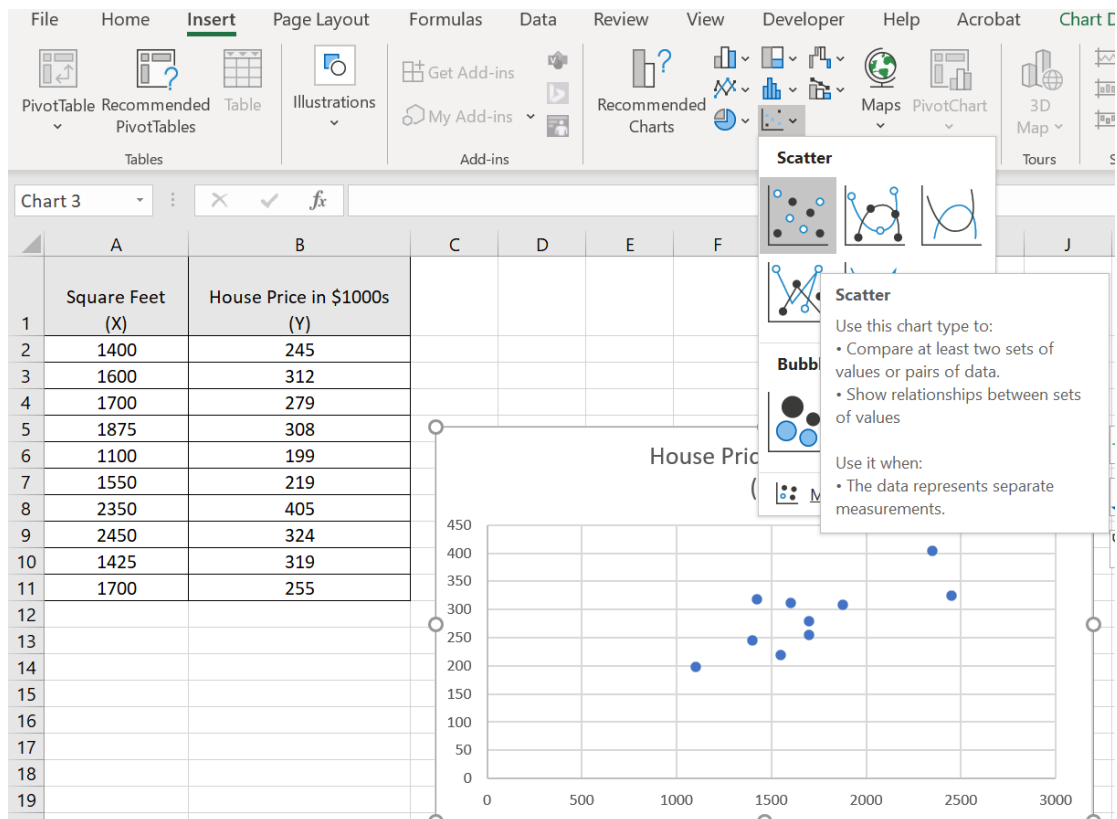
Plot a scatter chart and regression line.

Solution:

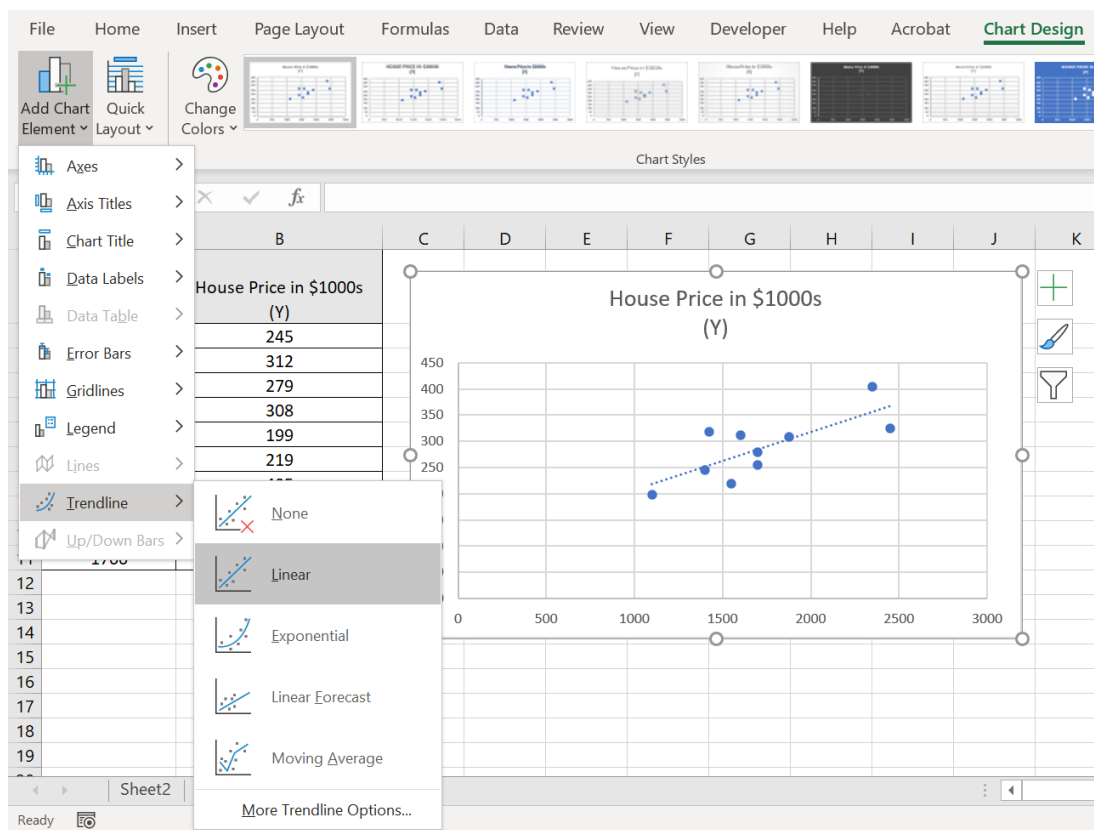
Open **Topic 7 Workbook.xlsx** and save file as **Topic 7 Workbook {Admin No}.xlsx**

Enter the data into Cell A1 to B11 as shown below.

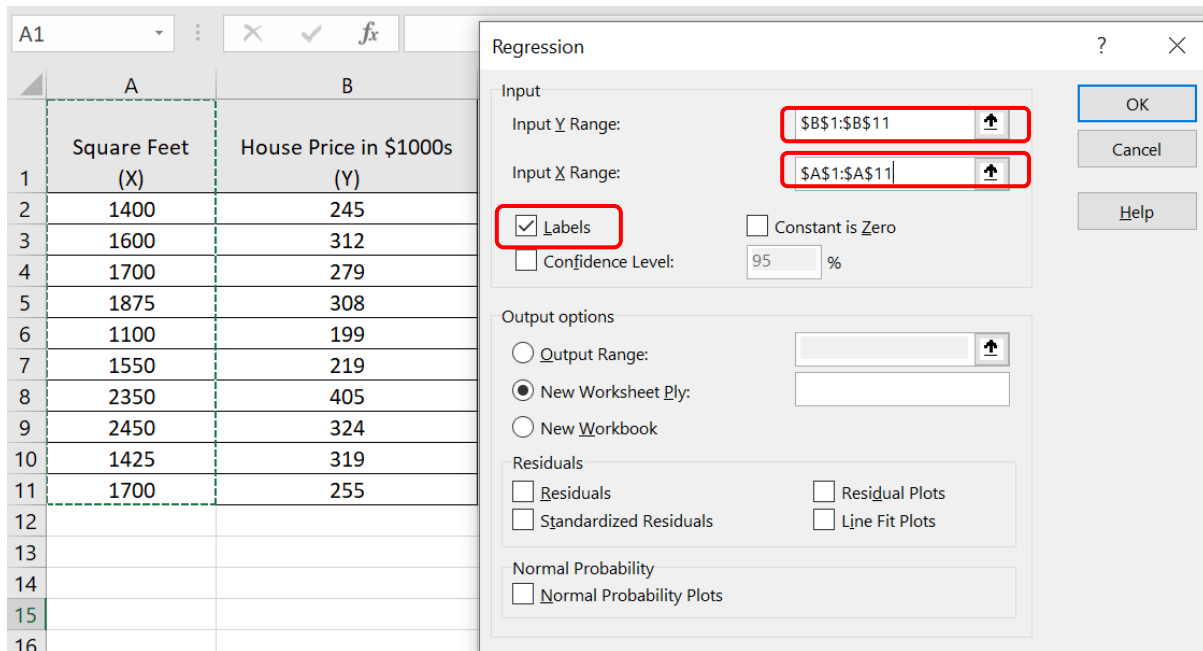
1. Highlight data values from A1 to B11.
2. On Insert tab, select Scatter Chart. You may use the chart tools to add titles or to do adjustment of other settings.



- Click anywhere on the chart. Click on **Chart Design** tab -> **Add Chart Element** -> **Trendline** -> **Linear**. A linear regression line is drawn.



4. Select Data tab -> Data Analysis -> Regression.
5. Input the Y range by selecting B1:B11
6. Input the X range by selecting A1:A11
7. Select the Labels checkbox.



Excel Output:

SUMMARY OUTPUT						
Regression Statistics						
Multiple R	0.762113713					
R Square	0.580817312					
Adjusted R Square	0.528419476					
Standard Error	41.33032365					
Observations	10					
ANOVA						
	df	SS	MS	F	Significance F	
Regression	1	18934.93478	18934.93478	11.08475762	0.010394016	
Residual	8	13665.56522	1708.195653			
Total	9	32600.5				
	Coefficients	Standard Error	t Stat	P-value	Lower 95%	Upper 95%
Intercept	98.24832962	58.03347858	1.692959513	0.128918816	-35.57711198	232.0737712
Square Feet	0.109767738	0.032969443	3.329377962	0.010394016	0.033740065	0.18579541

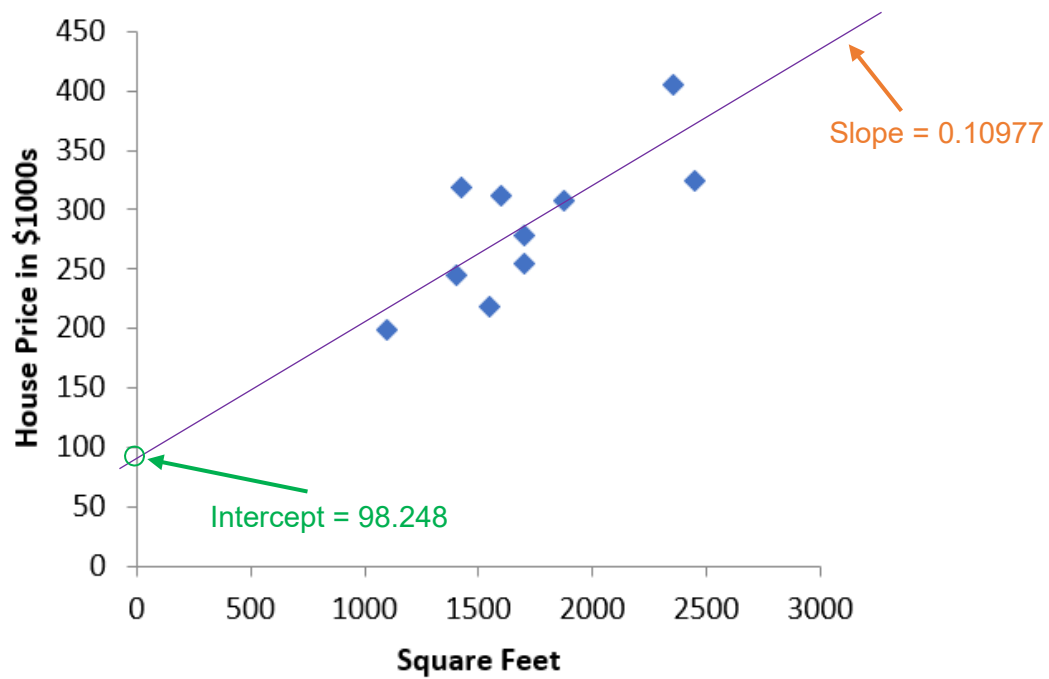
Regression equation is:

Using $y = mx + c$

House Price = 0.1097 (Square Feet) + 98.2483

Graphical presentation:

- Scatter plot and regression line



Tutorial 7

Exercise 1

The table below shows the Polytechnic GPA and University GPA of 10 students.

Polytechnic GPA, x	University GPA, Y
2.7	2.2
3.1	2.8
2.1	2.4
3.2	3.8
2.4	1.9
3.4	3.5
2.6	3.1
2	1.4
3.1	3.4
2.5	2.5

Using Excel, plot a scatter chart and regression line.

- a) From the Excel output,
 - i. State the correlation coefficient and the relationship between the two variables
 - ii. Write the equation of the regression line Y on X
- b) Using the equation of the line in part aii), find the University GPA if the Polytechnic GPA is 3.6.
- c) Using the line of best fit in part aii), find the Polytechnic GPA if the University GPA is 2.3.